

```

library(ape)
library(dplyr)

##
## Attaching package: 'dplyr'
## The following object is masked from 'package:ape':
##
##      where
## The following objects are masked from 'package:stats':
##
##      filter, lag
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
data(carnivora)

x <- c(1:8, NA, 10)

(
  d <- carnivora %>%
    select(SuperFamily, SW) %>%
    na.omit() %>%
    group_by(SuperFamily) %>%
    summarise(n = n(), v = var(SW), xbar = mean(SW))
)

## # A tibble: 2 x 4
##   SuperFamily      n      v xbar
##   <fct>          <int> <dbl> <dbl>
## 1 Caniformia      57 4232.  23.4
## 2 Feliformia      55 1099.  16.6

```

Example

```

alpha <- 0.05
xbarC <- d$xbar[1]
xbarF <- d$xbar[2]
diff <- (xbarC - xbarF)
# difference of means
nC <- d$n[1]
nF <- d$n[2]
vC <- d$v[1]
vF <- d$v[2]
t.star <- qt(1 - alpha / 2, df = min(c(nC, nF) - 1))
se <- sqrt(vC / nC + vF / nF)
ME <- t.star * se
(ci <- c(diff - ME, diff + ME))

## [1] -12.61944  26.30350

```